Sequence-Based Prediction of Single-Residue Mutation Effects on PSI in Degron Assays and Development of an Analysis Tool

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CT553H · Research Progress Report · 7 November 2026

From biology to prediction evidence

Full research report; detailed result tables are retained in the appendix

01 **Biological foundations and prediction task**
Protein → UPS → degron → mutation → ΔPSI

02 **Related work and experimental data**
Assay, provenance, quality control and observation units

03 **Methods and evaluation design**
Ridge, pLMs, CNN/mCNN, LoRA and leakage controls

04 **Available experimental results**
Historical H-sat; full-PSI W; scanning T; OOF replay

05 **Provisional findings and next steps**
Supported claims, limitations and unresolved evaluations

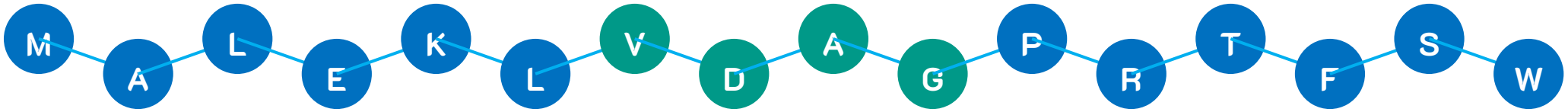
06 **Appendix and verifiable sources**
All 88 v3 recipes; historical controls; references

Proteins: amino-acid chains with cellular functions

A · Biological foundations

Sources: [P1] · [P11]

20 amino-acid types → sequence → structure / interactions → function



Protein

A molecule with functions in the cell.

Peptide

A short chain segment; the core data use 28 AA.

Residue

One amino acid at one position in a chain.

Stability must be defined by its measurement

A · Three related concepts describe different quantities

Sources: [P11] · [P10] · [P1]

Protein abundance

The amount of protein at a given time depends on both synthesis and degradation.

Protein lifetime

Turnover and half-life describe replacement or degradation over time.

Structural stability

Thermodynamic stability concerns protein folding; it differs from reporter PSI.

UPS: recognition, ubiquitination and degradation

A · Ubiquitin–Proteasome System

Sources: [P1] · [P11]

E1

Activates ubiquitin (Ub).

E2

Carries Ub to the tagging machinery.

E3 + substrate

Recognizes the degron / compatible context.

Proteasome

Recognizes tagged protein and degrades it.

Degrons contribute to recognition; ΔPSI prediction does not identify an E3.

A degron acts within a protein context

A · A sequence segment can contribute to targeting a protein for degradation

Sources: [P1] · [P2]

THE SAME SEQUENCE SEGMENT

In the native protein

May be buried, masked by interactions or dependent on post-translational modifications.

In the peptide reporter

The 28-AA segment is placed in a construct; context and exposure may differ from the native protein.

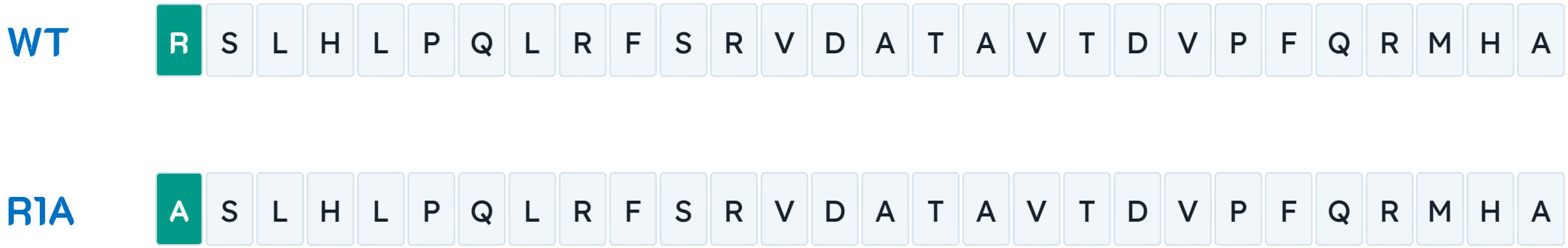
Reporter effects therefore provide assay-specific evidence.

A single substitution changes one residue

A · Wild type (WT): reference sequence; mutant: sequence after substitution

Sources: [D1] · [P1]

A2RUQ5_C17orf102_2



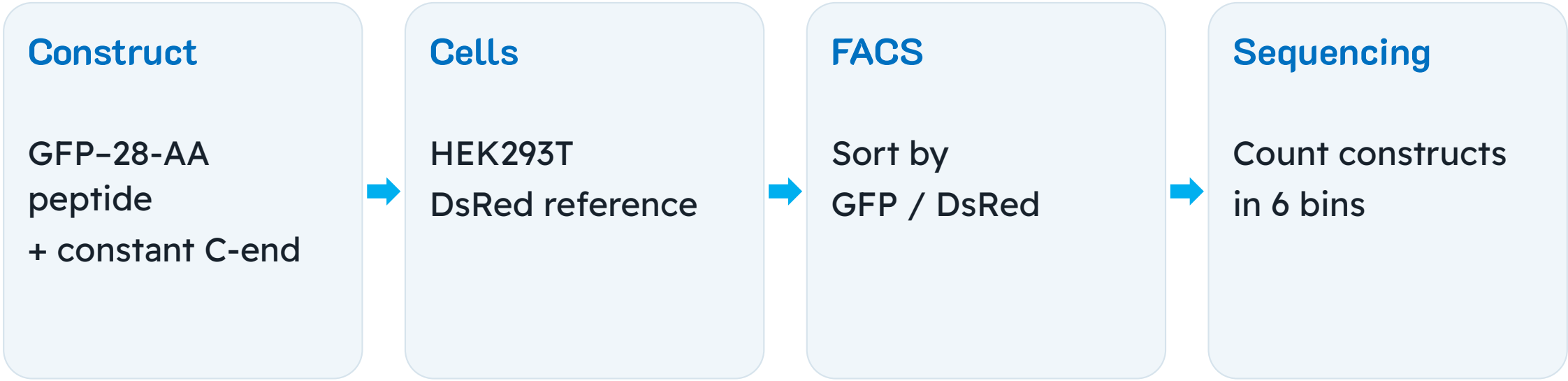
One substitution at site 1: Arginine (R) → Alanine (A).

Recognition or context may change; measurements establish the effect.

The GPS assay turns reporter behavior into data

A · HEK293T; GFP–28-AA peptide reporter; DsRed reference; six FACS bins

Sources: [P1] · [P10] · [D4]



The measurement is specific to the reporter, construct and cellular conditions.

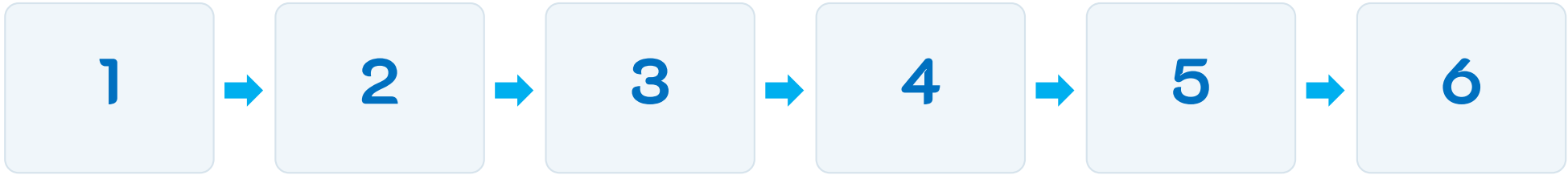
PSI is a reporter index on a scale from 1 to 6

A · Protein Stability Index in the Zhang assay; not a time unit

Sources: [P1] · [D4]

$$PSI = 1 \times p(1) + 2 \times p(2) + \ldots + 6 \times p(6)$$

p(b): bin-normalized read proportion; the proportions sum to 1.



Low PSI: less stable reporter

High PSI: more stable reporter

WT and mutant measurements define ΔPSI

A → B · Real R1A record; displayed values are rounded

Sources: [D1] · [D3]

PSI(WT)

4.0428

Median PSI of synonymous variants per half.

PSI(mutant)

4.4232

Mean of sat_a and sat_b.

ΔPSI

+0.3804

Higher reporter stability within this assay.

$$\Delta = \tfrac{1}{2}\left[(PSI_mut_a - WT_a) + (PSI_mut_b - WT_b)\right]$$

The sign of Δ does not establish degron disruption or a specific E3 mechanism.

The task is mutation-effect regression

B · This endpoint differs from degron / non-degron classification

Sources: [M1] · [M2] · [D1]

Input x

WT sequence of 28 AA, site i and replacement amino acid a. Exactly one substitution generates the mutant.

Target y

f(x) ≈ ΔPSI = PSI(mutant) − PSI(WT).

Δ > 0: higher reporter stability; Δ < 0: lower reporter stability.

Scope

Sequence-only input; single substitutions in peptide reporters. Measured PSI, read counts and fold IDs are not model features.

Five questions organize the progress report

B · Current organizing questions; the complete history was not preregistered

Sources: [M1] · [K1] · [L1] · [U2]

RQ	Intervention to evaluate	Control / evidence
1	Does broader training PSI coverage help?	Ridge B/D/E; identical W test population
2	Do representations / positional mixing help?	one-hot ↔ ESM; k1 ↔ CNN ↔ mCNN
3	Does scanning help saturation prediction?	sat-only ↔ pooling / pretrain
4	Does backbone adaptation help?	frozen ↔ LoRA matched B/D
5	Does performance transfer or support prioritization?	grouped OOF; calibration; T; replay k10

Prior work already addresses several endpoints

C · Compare task, data and exposure; published scores use different protocols

Sources: [P1] · [P2] · [P3] · [P14] · [H1] · [H3]

Source	Data / method	Contribution and relevance to CT553H
Zhang 2023 [P1]	GPS 28-AA; scan + saturation; DegronID	Measured footprints / E3 pairs; primary label source, not CT553H model performance
Voutsinos 2026 [P2]	30-AA peptides; CNN-based PAP; ΔPAP	Abundance prediction and variant follow-up; ΔPAP is not automatically calibrated in ΔPSI units
Degformer [P14]	Transformer PSI prediction, including mutant mode	Training exposure must be audited; the public checkpoint overlaps the benchmark
Zheng 2024 [P3]	MetaDegron: pLM with multimodal features	E3-targeted degron prediction; endpoint differs from substitution ΔPSI

Method choices are hypotheses to test

C · A scoped evaluation question; absolute novelty is not established

Sources: [P4] · [P5] · [P6] · [P7] · [P9]

Source / approach	Core idea	Bounded use in this study
Hoerl–Kennard 1970 [P7]	Ridge: linear regression with an L2 penalty	Simple control to assess representation value
Lin 2023 [P5]	ESM-2 learns protein-sequence representations	Frozen embeddings do not guarantee better ΔPSI prediction
Hu 2022 [P6]	LoRA learns low-rank updates to a frozen base	Test rank-8 Q/V adaptation; NLP resource savings do not transfer automatically
Szulc 2024 [P4]	DEGRONOPEDIA inspects motifs / context	Biological case evidence; not the same regression benchmark
Larsen 2025 [P9]	Degron mapping in transcription factors	Preprint / potential external source; not included in the benchmark

From original sources to the v3 data bundle

D · Mendeley v2 / GSE240610 → PSI / QC → tables → splits

Sources: [P1] · [P15] · [D2] · [D4]

Zhang sources

Mendeley v2
CC BY 4.0
GSE240610

Reconstruct PSI

Read 6 bins
normalize → PSI
QC + WT

Bundle v3

Tables / provenance
122,412 sat cells
224,470 scan cells

Evaluation

Group splits
fit / OOF / T
artifact + hash

Saturation covers 19 replacements; scanning is sparse

D · A mutation cell is a site × amino-acid observation, not a biological cell

Sources: [P1] · [D2] · [M1]

Saturation

19 non-WT amino acids per site.

Up to 28 × 19 = 532 cells/peptide.

235 peptides; 122,412 QC-passing cells.

Primary target labels on W.

Scanning

Rule-based R/A/S substitution per site.

More peptides, fewer replacement types.

224,470 cells in the replicate union.

dev for training; T held out.

Full-PSI training retains technical quality control

D · Missing cells are not imputed; test data do not fit quality weights

Sources: [D2] · [D3] · [M1]

Reads / sequence

Reads ≥10

28 standard AA

Valid mapping

Source-specific WT

Sat: median

≥10 synonymous cells

Scan: _mut0

Match observations

Sat: both halves

Scan: retain masks rep1 / rep2

Retain provenance

Missing ≠ 0

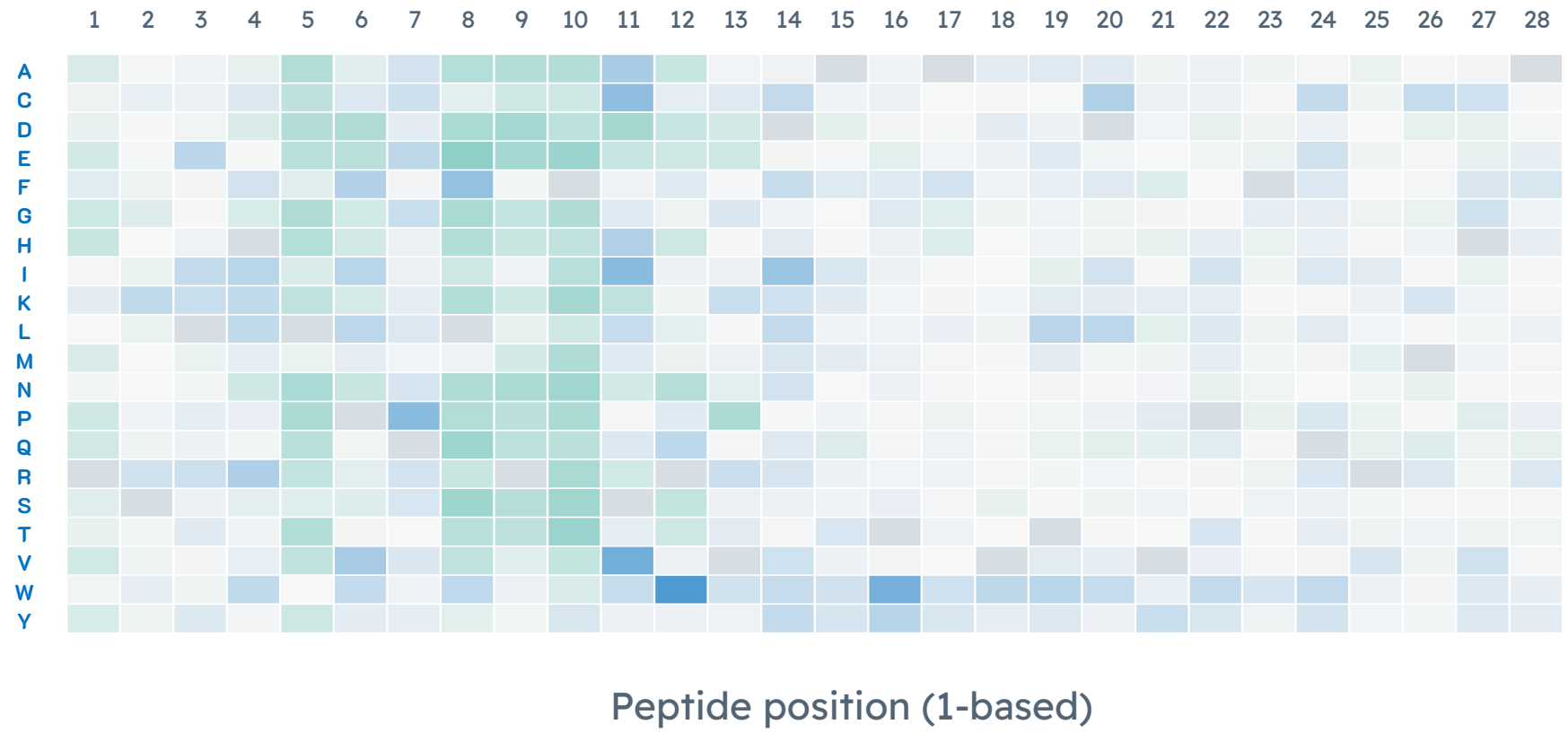
No label imputation

PSI is metadata

A measured mutation matrix for one peptide

D · A2RUQ5_C17orf102_2; color shows measured ΔPSI; gray marks WT cells

Sources: [D1]



Measured ΔPSI

Blue: negative

White: near 0

Teal: positive

Gray: WT / missing

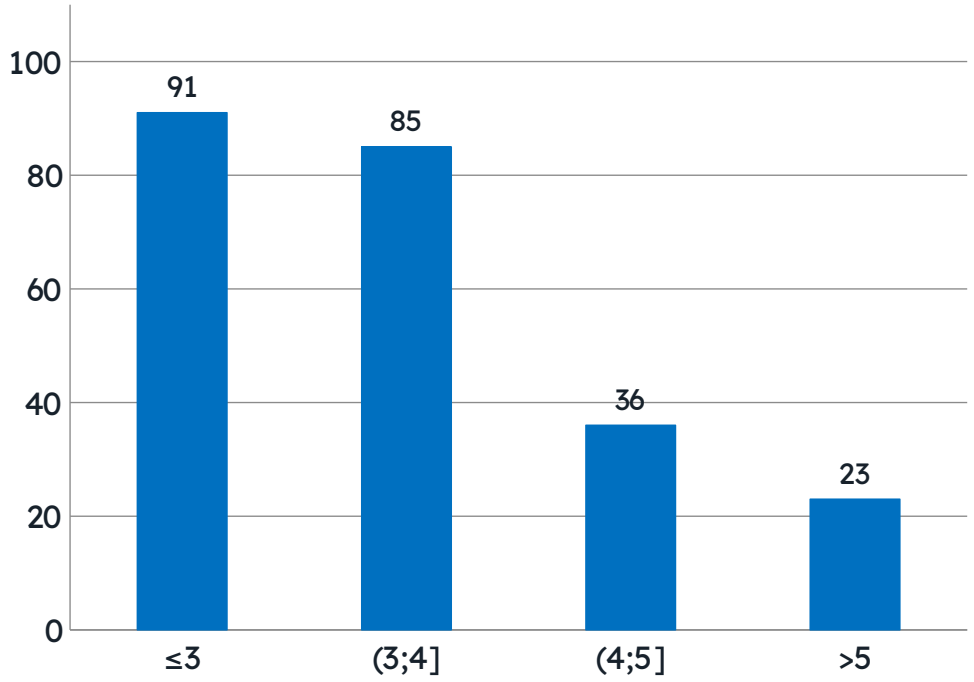
Display scale –3 to +3.

Full-PSI W: 235 peptides, 122,412 mutations

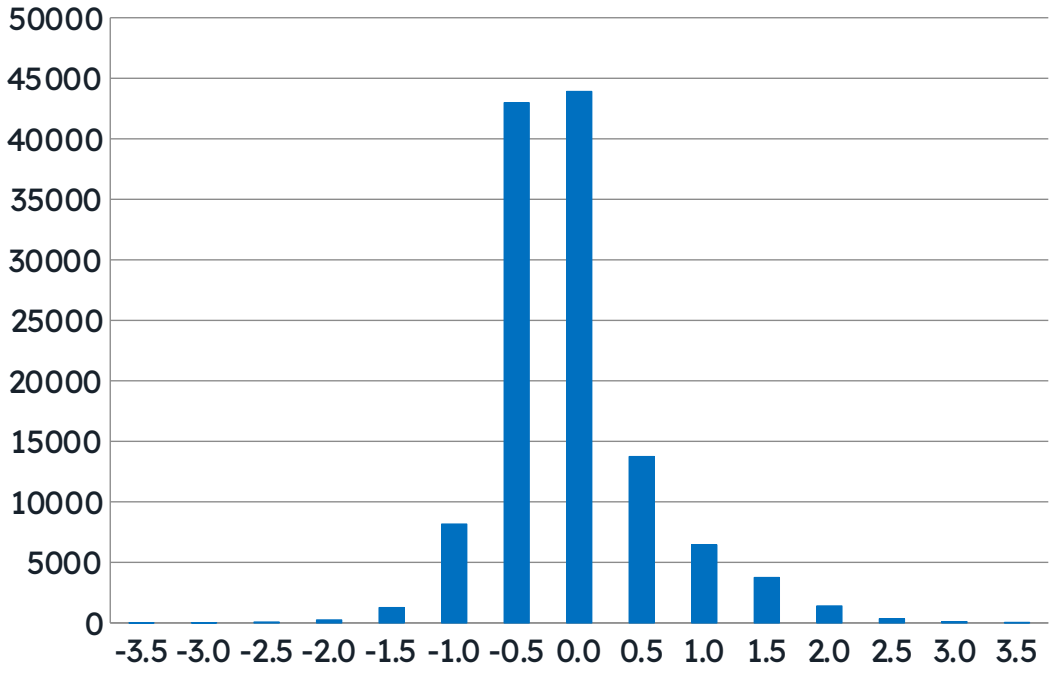
D · 221 protein groups; 12,053 cells with Δ≥1 and 1,577 with Δ≤−1

Sources: [D1] · [D2]

Peptides by WT PSI



Measured ΔPSI distribution: 122,412 cells



Measurement quality and coverage bound inference

D · Ranking metrics use their own eligible-peptide populations

Sources: [D1] · [K1] · [M1]

Missing measurements

Not every peptide has all 532 cells. Use QC-passing observations; a missing cell does not mean Δ = 0.

Replicate differences

scan_dev has 152,004 paired rep1/rep2 cells; Pearson r = 0.7074. Disagreement is a quality indicator, not a known variance.

Assay floor / ceiling

$1 - PSI(WT) \leq \Delta PSI \leq 6 - PSI(WT)$. WT PSI >5 cannot support a positive event $\Delta \geq 1$.

Protein-group splitting reduces leakage risks

D · cov80 + bidirectional cov50 protein hits + exact-sequence unions

Sources: [D2] · [M1] · [P8] · [P12]

Protein hits

cov80 + cov50
identity ≥30%
alignment ≥50

Group union

Exact sequences
+ homology edges
221 sat groups

5 outer folds

One fold per group
47 peptides/fold



Tune inside; evaluate out of fold outside

D / F · 5 outer folds; inner validation ≈20% of groups; seeds 17, 29, 43

Sources: [M1] · [L2]

Outer train

Grouped train / val.

Fit scaler/PCA on train.
MSE selects α / epochs.

Refit the recipe

Reinitialize the neural model.

Fit eligible outer-train for the selected epochs.

Outer test → OOF

No tuning here.

Repeat 5 folds × 3 seeds; ensemble predictions.

Calibration also uses its own fold validation; global OOF fitting can leak labels.

Five training arms share the full W test set

D / F · Arms restrict training only; B still uses high-PSI inner validation

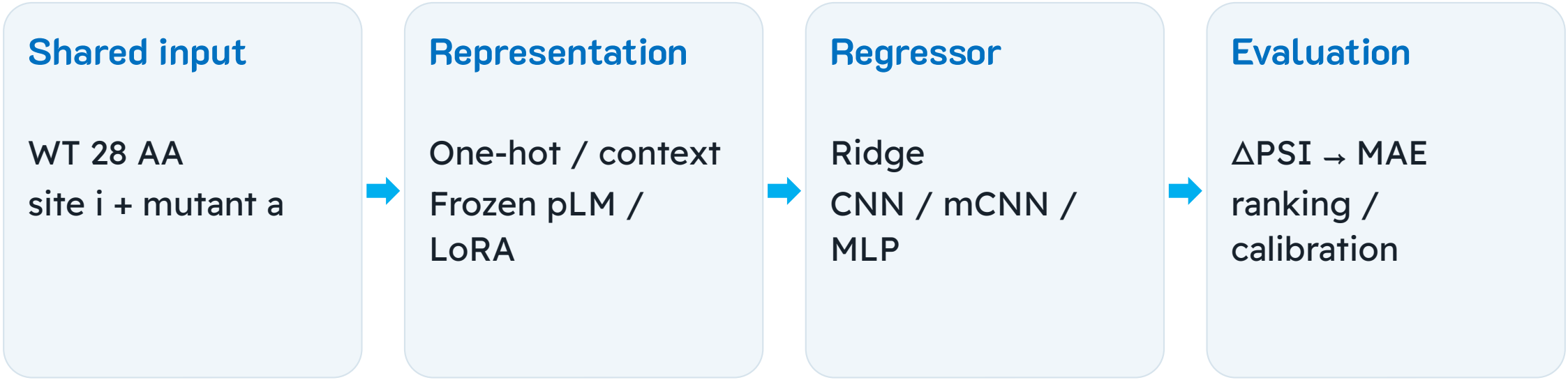
Sources: [M1] · [D2]

Arm	Train saturation	Validation / test	Question
A	PSI(WT) ≤3	Full range	Historical domain
B	PSI(WT) ≤4	Full range	Primary reference
C	PSI(WT) ≤5	Full range	Wider coverage
D	Full range	Full range	Broader data
E	Full range; balance stratum → peptide	Full range	Coverage + weighting

One data contract, multiple representations

E · Data → features / encoder → regression → evaluation → calibration

Sources: [M1] · [M2] · [M3] · [M4]



Shared labels and W test set; vary components within their matched comparisons.

Ridge W5 is an ordered-context baseline

$E \cdot x \in R^{260} \rightarrow$ scaling $\rightarrow$ Ridge $\rightarrow \Delta PSI$

Sources: [M2] · [M1] · [P7]

Pair

WT one-hot 20
Mutant one-hot 20
= 40 dimensions

Composition

Whole-WT amino-acid frequencies
= 20 dimensions

Ordered W5

10 neighboring sites
× 20 AA
= 200 dimensions

Ridge

Concatenate 260
Scaling + L2 penalty
→ ΔPSI

Controls: pair = 40; pair+composition = 60; unordered window = 80 dimensions.

Frozen pLMs represent the WT-to-mutant change

E · ESM2-35M: 28 × 480; Δsite + meanΔ gives 960 dimensions

Sources: [M1] · [P5] · [M2]

WT / mutant

Shared encoder
28 × 480
ESM2-35M

Hidden difference

ΔH = H_mut − H_WT
Site + mean
= 960 dimensions

Frozen Ridge

PCA64 on Δ
+ W5 context 260
→ regression

Fit PCA on train only; retain the context block. Frozen encoders do not learn ΔPSI labels.

CNNs learn from a crop around the mutation site

E · ESM crop ±7: 15 × 480; all modules are verified against code

Sources: [M3] · [M1]

Input crop

WT one-hot 20
+ center flag 1
+ Δhidden 480

Projection

480 → 32
15 × (20+1+32)
= 15 × 53

Conv k5

53 → 96 channels
ReLU + mask
15 positions

Pool + MLP

Max96 + center96
+ auxiliary60
252 → 64 → 1

Extract Δhidden from both WT and mutant; auxiliary = pair40 + composition20.

Positional-mixing ablation: k1, k5 and multi-scale

E / F · Shared crop, auxiliary inputs and splits; parameter counts differ

Sources: [M3] · [M1] · [K2]

Head	Kernel / filters	After pooling	Hypothesis
Pointwise	k1 ×96	96 max + 96 center	No positional mixing in the convolution
CNN	k5 ×96	96 max + 96 center	Local context helps prediction
mCNN	k3 / k5 / k7; 32 filters per branch	3 × (32 max + 32 center)	Multiple scales may add information

LoRA updates Q/V while preserving base weights

E · ESM2-35M; rank 8, alpha 16; frozen and LoRA share the head design

Sources: [M4] · [M1] · [P6]

$$W' = W + (alpha / r) \times B \times A$$

Frozen base

ESM2-35M
W stays fixed

Adapter Q/V

r = 8; alpha = 16
Learn A and B

Mutant – WT

Δsite480
+ meanΔ480
+ auxiliary60

Matched MLP

1,020 → 64 → 1
Matched head init
Frozen ↔ LoRA

Scanning supports transfer; T stays outside fitting

E · scan_dev → 5 pretraining epochs → saturation fine-tuning; scratch control

Sources: [M1] · [M3] · [D2]

scan_dev

5,489 peptides
rep1 / mean12
source train

Pretrain

One-hot CNN
5 fixed epochs
Exclude T

Fine-tune W

Saturation
CNN scratch control
Same architecture

scan_T

Held-source
1,410 scored
W5-D only

Keep 1,278 peptides overlapping saturation groups outside source training.

Implemented comparisons retain matched conditions

F · OOF / post-hoc development analysis; run and seed details are in Notes

Sources: [M1] · [K1] · [L1] · [U2]

Experiment	Intervention	Fixed conditions / objective
Q1 / stage01	A–E; pair / unordered / ordered	W folds; full-range validation/test; primary MAE
Q2 / stage04–05	pLM family; head k1/k5/3,5,7	Matched head / feature conditions within each contrast
Q3 / stage06	sat-only / scan / quality / pretrain	Same W cohort; T excluded from fitting
Q4 / stage11 local	frozen ↔ LoRA; arms B and D	Same runtime, head initialization and splits
Q5 / stage07	Calibrate; lock W5-D before scoring T	Inner-only calibration; separate T cohort
Replay 07/11	LoRA-D ↔ W5-D ↔ random	10 selections/peptide; 2 primary contrasts

Three evaluation views answer different questions

F · MAE: 235 peptides; AUROC/AP+: 184, AUROC/AP−: 110

Sources: [M1] · [K1] · [U2]

MAE ↓

PSI units; peptide-macro averaging

How large are effect-size errors? Calculate MAE within each peptide, then average across peptides.

AUROC / AP ↑

Δ≥1 or Δ≤−1; at least 3 cells in each class

How well are events ranked? AP depends on event prevalence and is not accuracy.

Precision@10 ↑

Replay retains all 235 peptides

How many known hits are found in exactly 10 selections per peptide? Keep peptides with no hits.

Resample protein groups, not mutation cells

F · 5,000 paired bootstrap draws; ensemble predictions precede metric calculation

Sources: [M1] · [P13] · [L1] · [U2]

Prediction

For each cell
3 seeds → mean
Then score

Macro

MAE per peptide
→ average
across peptides

Bootstrap

Sample group_id
Keep its peptides
5,000 paired draws

Confidence interval

MAE difference <0 helps
AUROC difference >0 helps
Retain multiplicity

What evidence is available as of 7 November?

G · Saved 6 November audits; no new model training or scoring in this update

Sources: [K1] · [L1] · [L3] · [U1] · [S1]

Component	Status	Evidence scope
Local v3	633 completed jobs; 41 recipes	3 seeds; 5 outer folds; 122,412 cells/seed
Kaggle v3	47 recipes; separate snapshot	45 recipes with 3 seeds; 2 frozen recipes with seed17 only
Locked T	W5-D scored	39,234 cells / 1,410 peptides / 730 groups
Replay OOF	Measured on 7 November	235 peptides; equal budget: 2,350
External / structure	source_pending / structure_pending	No valid benchmark yet
G2 / serving / wet lab	Decision / implementation pending	Not inferred from development rankings

Historical results use the same H-sat test set

G · H-sat and full-PSI W differ in population, grouping and protocol

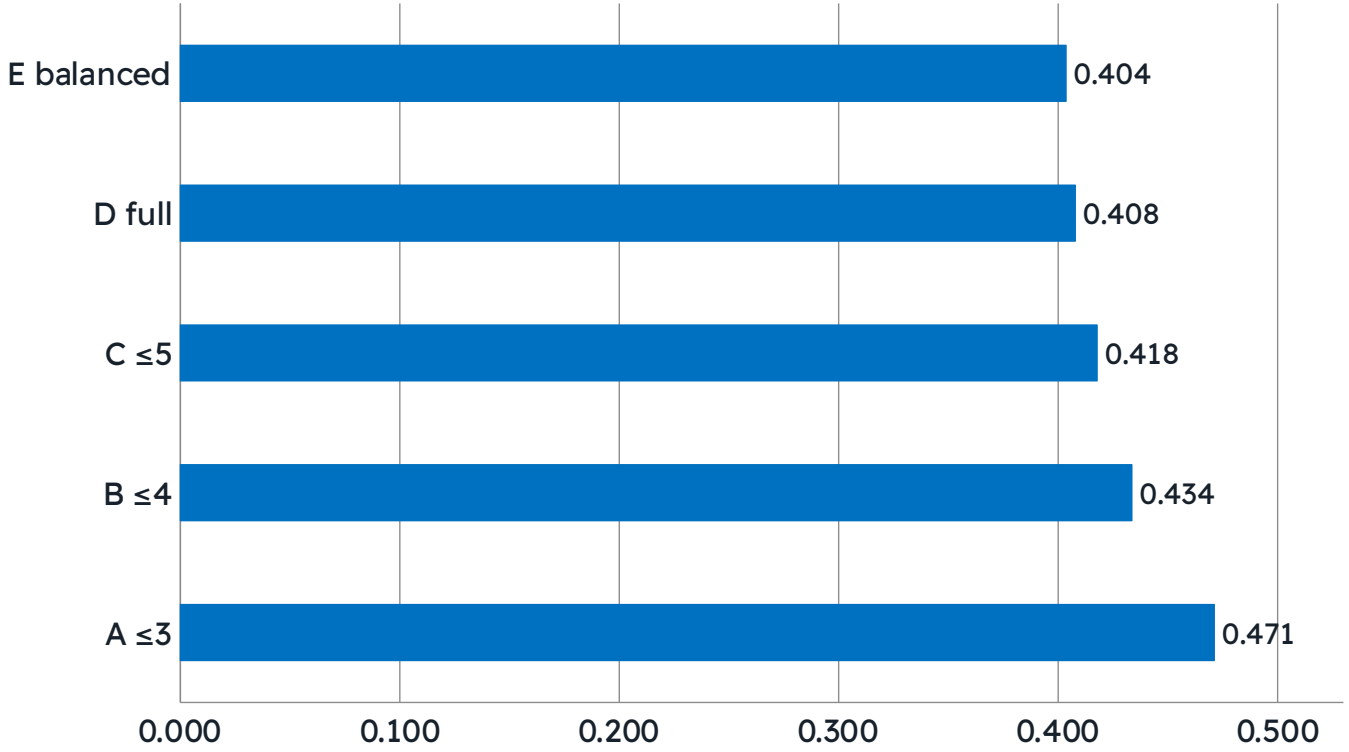
Sources: [H1] · [H2] · [H3] · [H5]

Recipe / control	AUROC+	Bounded interpretation
ΔPAP (published weights)	0.7693	Reference score; not a newly fitted matched-unit predictor
Ridge train WT PSI≤3	0.8082	91 peptides / 89 groups / 47,847 cells
Ridge train WT PSI≤4	0.8208	Δ +0.0126; 95% CI [+0.0061; +0.0190]
Degformer public checkpoint	0.9248	Training overlap; not a clean generalization comparator
Degformer cleaned training	0.8068	3 seeds ×5 folds; CI versus Ridge includes 0
R6 ESM35-mCNN / W5	0.8305 / 0.8286	mCNN–M3 primary contrast unresolved; G2 not closed

Broader training coverage improves Ridge W5

G · Identical W test set; peptide-macro MAE ↓; 3-seed prediction ensemble

Sources: [K1]



D versus B
−0.0258 PSI

E versus B
−0.0300 PSI

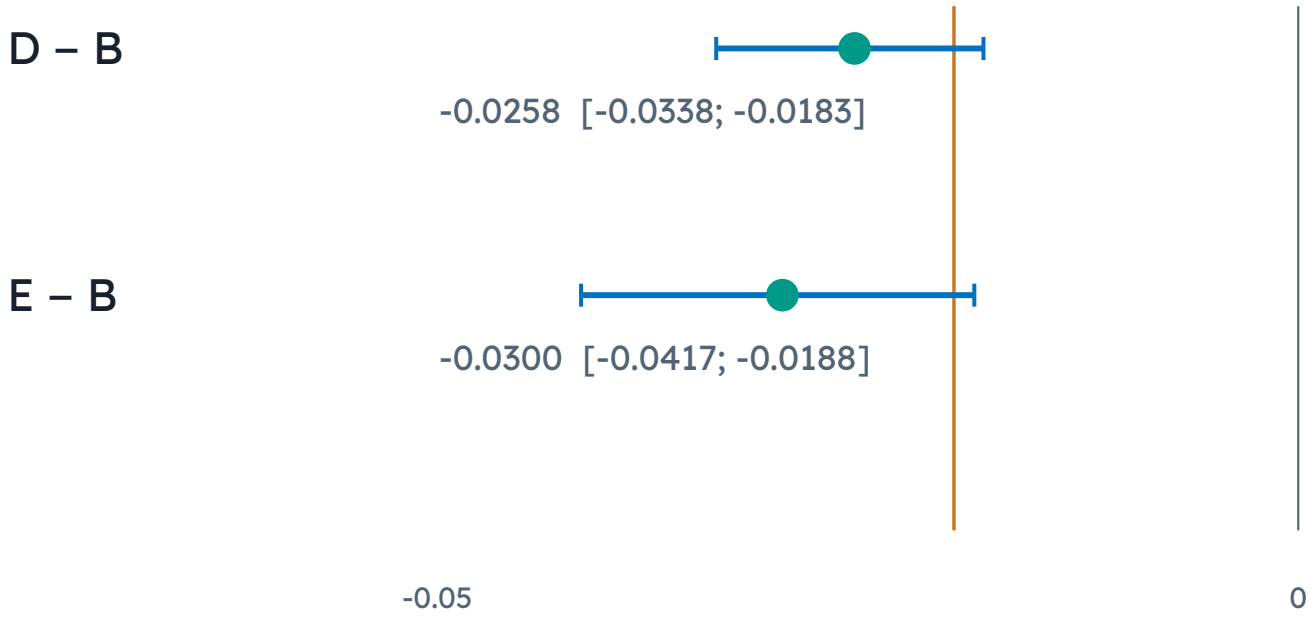
Larger gains at high WT PSI.

Q1 improves MAE; the practical margin is unresolved

G · 97.5% CIs for D–B/E–B; the –0.02 margin belongs to Q1 only

Sources: [K1] · [M1]

ΔMAE: lower is better



CIs below 0 support improvement.

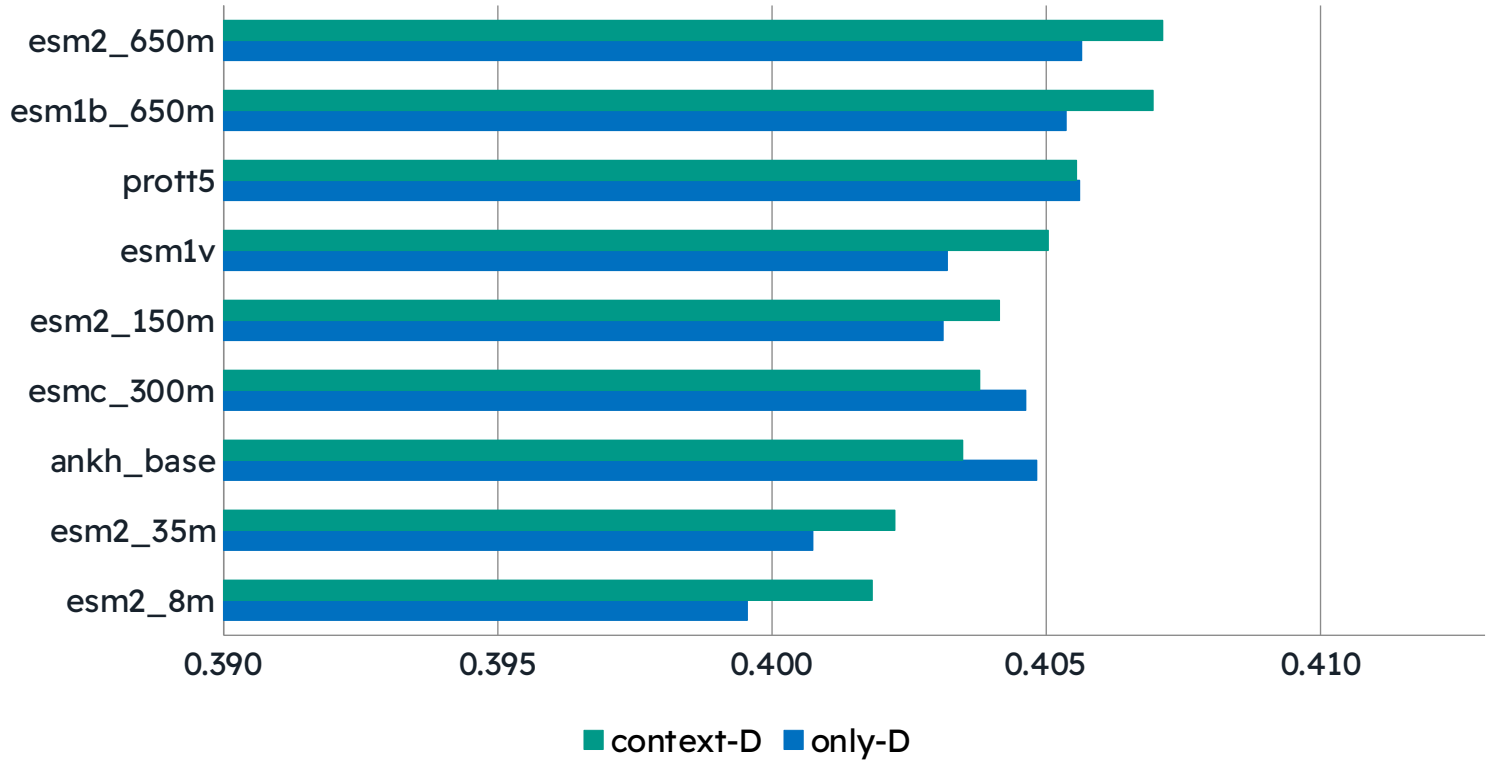
Not entirely below –0.02.

E does not establish the low-PSI AUROC+ guardrail.

Larger pLMs do not yield monotonically lower error

G · Frozen pLM + PCA64/Ridge; arm D; same local runtime

Sources: [L1]



9 evaluated checkpoints

Nested selection:

MAE 0.3945

AUROC+ 0.8409

The score evaluates inner-fold selection.

ESM CNN helps; mCNN superiority remains unresolved

G · Development; descriptive CIs; k1 and k5 heads differ in parameter count

Sources: [K1] · [K2]

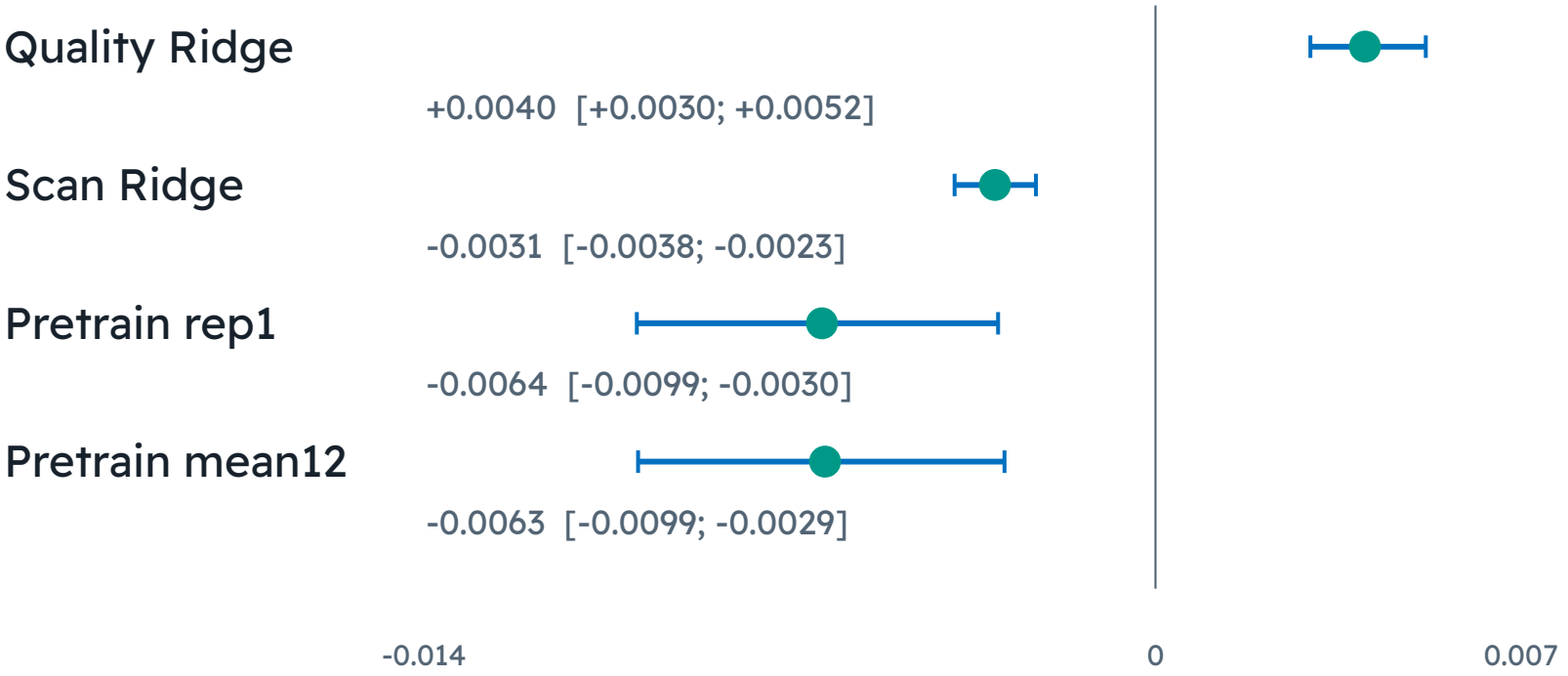
ESM2-35M / D	MAE ↓	AUROC+ ↑	Paired comparison
Pointwise k1	0.3897	0.8375	Reference without convolutional positional mixing
CNN k5	0.3844	0.8488	ΔMAE −0.00534; 95% CI [−0.01017; −0.00090]
mCNN k3/5/7	0.3850	0.8463	mCNN−CNN: CI95 MAE [−0.00171; +0.00307]
mCNN / E (different arm)	0.3838	0.8470	mCNN-E−CNN-D: CI95 [−0.00316; +0.00222]

Scanning pretraining helps its tested control

G · One-hot CNN; same W; descriptive 95% CIs; quality weighting worsens MAE

Sources: [K1] · [K2]

ΔMAE: lower is better



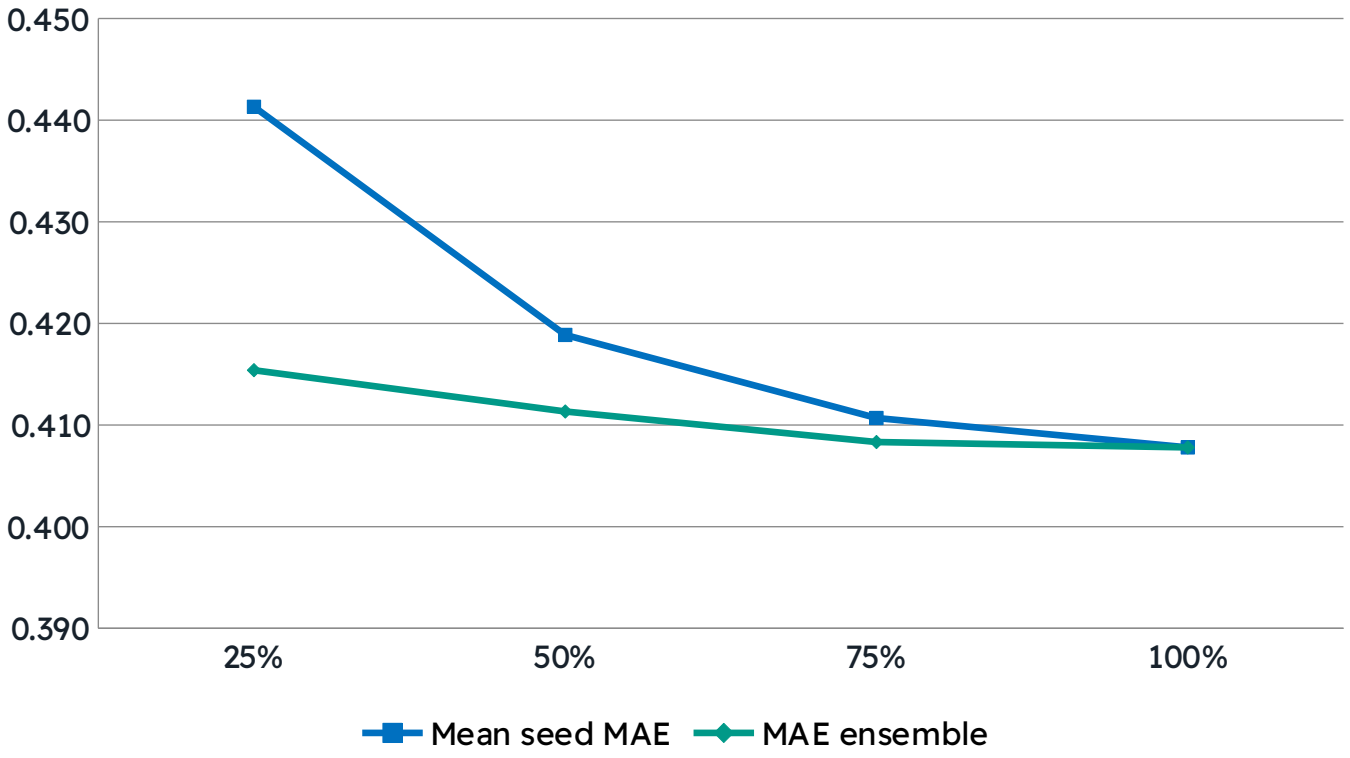
Quality weighting worsens MAE.

Rep1 and mean12 both help versus scratch; their difference is unresolved.

Learning curves: single fits versus ensembles

G · Ridge W5-D; 25/50/75/100% of training groups; fixed W test set

Sources: [K2]



Seeds use different subsets below 100%.

A 25% ensemble may see a union larger than 25%.

No equivalent LoRA sample-size curve yet.

LoRA improves matched frozen models in B and D

G · Matched runtime / head / split; three-seed prediction ensemble

Sources: [L1] · [M4]

ESM2-35M local	MAE ↓	AUROC+ ↑	AP+ ↑	AUROC− ↑
frozen-B	0.4034	0.8263	0.4554	0.7541
lora-B	0.3637	0.8550	0.5517	0.7921
frozen-D	0.3904	0.8246	0.4544	0.7667
lora-D	0.3650	0.8536	0.5474	0.8028

Group-level intervals support the LoRA gains

G · LoRA – frozen; 98.75% post-hoc Bonferroni CIs across four contrasts

Sources: [L1]

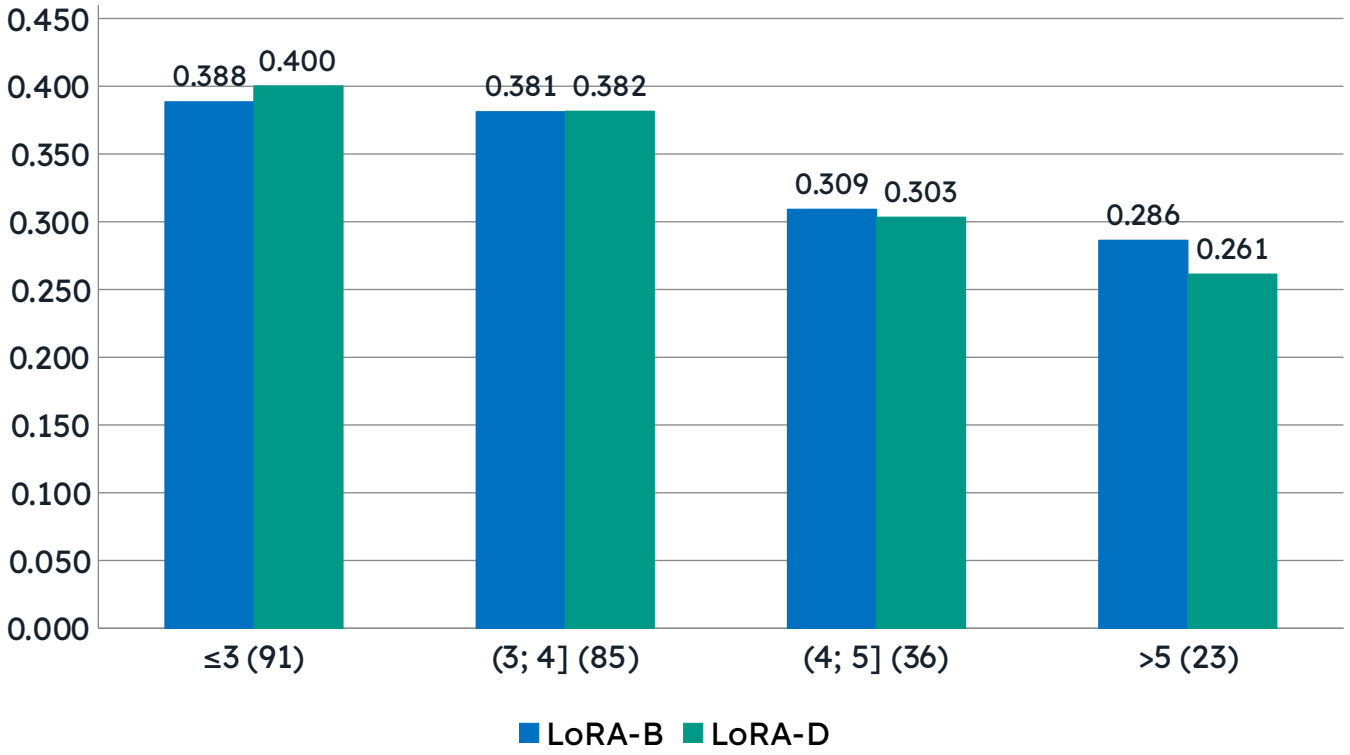


Matched adaptation on development W; LoRA transfer beyond the source remains untested.

LoRA-B and D trade off across WT-PSI strata

G · Whole-W D−B MAE = +0.00133; 95% CI [−0.00295; +0.00554]

Sources: [L1]



B has lower MAE at low WT PSI.

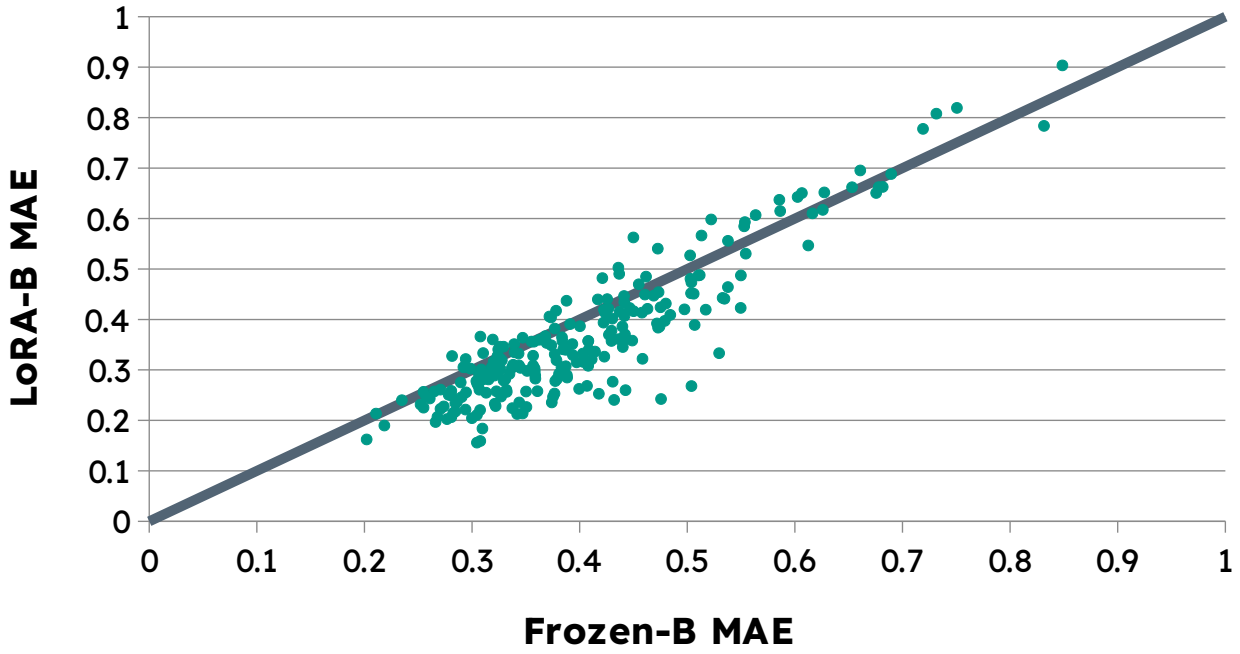
D has lower MAE in the two high-PSI strata.

Similar overall scores can hide different behavior.

Average gains coexist with peptide-level failures

G · LoRA-B improves 184/235 peptides; LoRA-D improves 164/235

Sources: [L2]



One point per peptide.

Below y=x favors LoRA.

B worsens 51 peptides.

D worsens 71 peptides.

CCDC17 tile9: LoRA-B MAE 0.9035 remains a failure case.

Large effects still have higher error and shrinkage; residuals do not establish a mechanism.

Lower Brier does not improve every calibration metric

G · Three-class calibration CV; inner-only; no final LoRA serving model

Sources: [L1] · [M1]

Local recipe	Brier3 ↓	ECE+ ↓
frozen-B	0.1789	0.0129
frozen-D	0.1795	0.0114
lora-B	0.1631	0.0163
lora-D	0.1659	0.0165

LoRA has lower Brier but higher ECE+ than frozen.

Inner-prior Brier3 = 0.2006.

Calibration CV is not a final export; model-card provenance remains incomplete.

Cell-micro; classes Δ≤−1 / neutral / Δ≥1. Different from binary P6 Brier.

Locked T: W5-D only; MAE gain remains unresolved

G · 39,234 cells / 1,410 peptides / 730 groups; rep1; R/A/S mutants

Sources: [L1]

T metric	W5-D	Condition
MAE macro	0.3991	1,410 peptides
MAE zero baseline	0.4024	Same cohort
ΔMAE / CI95	-0.0033	[-0.0092; +0.0028]
AUROC+ / AUROC−	0.8405 / 0.5176	338 / 85 eligible peptides
P(positive) / observed rate	0.0895 / 0.0512	Cell-micro; probability shift is present

OOF replay uses an equal mutation-selection budget

F / G · Primary: LoRA-D ↔ Ridge W5-D ↔ random; k = 10 per peptide

Sources: [U1] · [U2]

OOF per cell

W full-PSI

235 peptides

3 seeds → mean

Rank within peptide

LoRA-D / W5-D

Expected random

216–532 candidates

Select exactly 10

10 ×235

=2,350 substitutions

For every policy

Count known hits

Measured Δ≥1

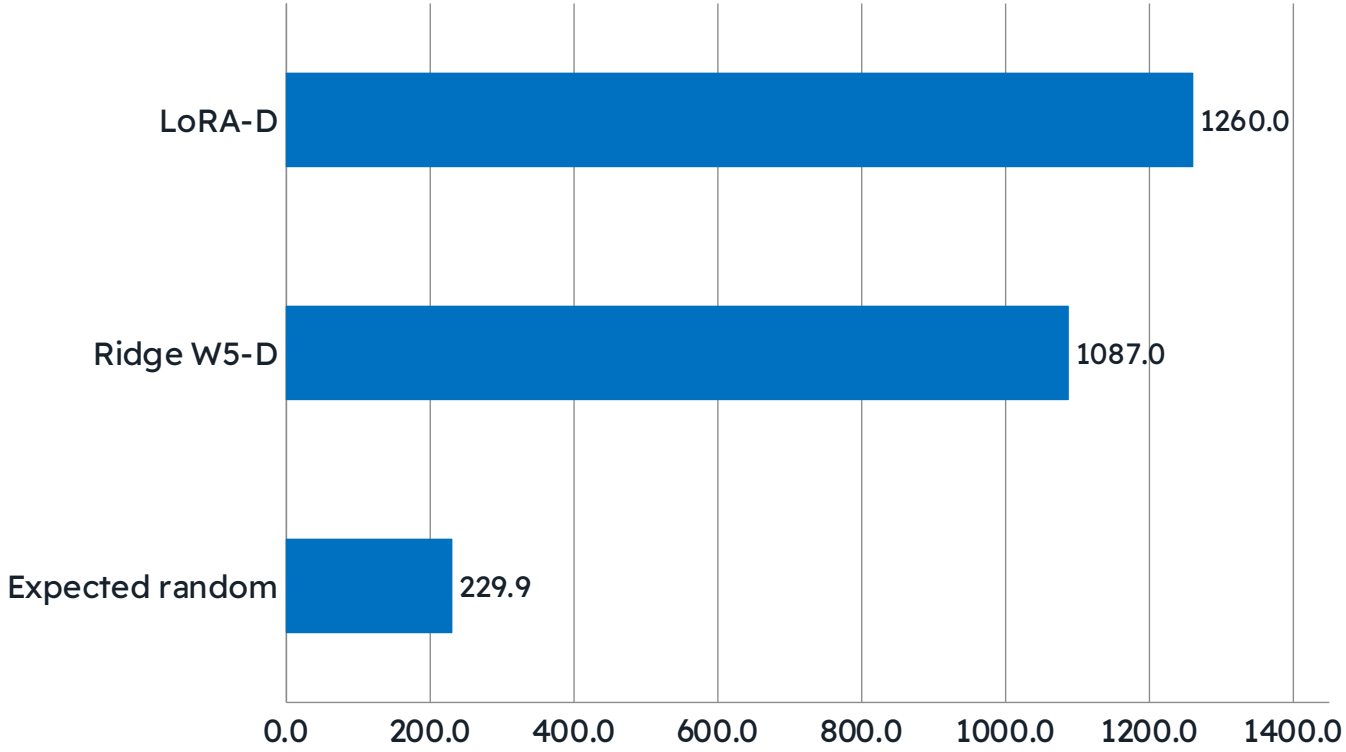
P@10: all 235

Recall:198 with hits

LoRA-D recovers 173 more hits at the same budget

G · Positive hit: measured ΔPSI≥1; 2,350 selections per policy

Sources: [U1]



Precision@10

LoRA-D 53.62%

W5-D 46.26%

Random 9.78%

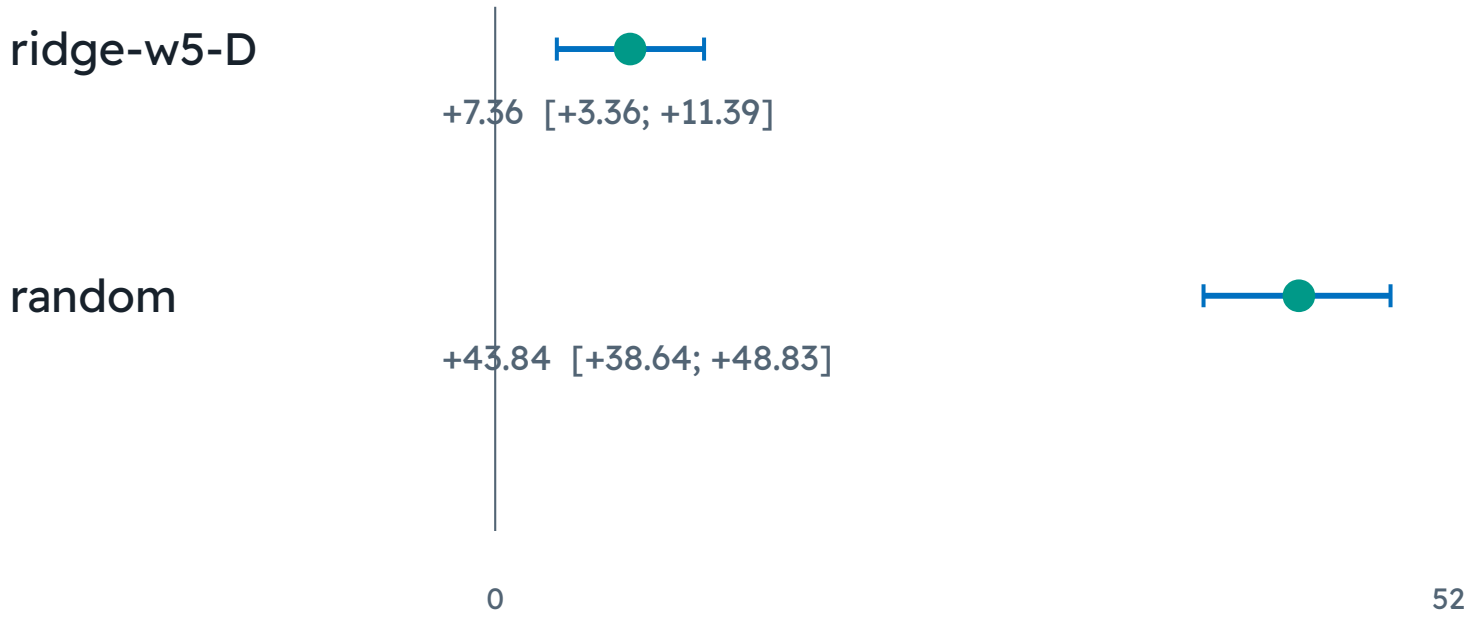
+173 hits versus W5-D in OOF replay.

The replay supports a prioritization benefit

G · ΔPrecision@10 in percentage points; 97.5% CIs for two primary contrasts

Sources: [U1]

LoRA-D minus baseline: percentage points



Both CIs are above 0.

To recover 50% of known hits:

71.32 selections/peptide for LoRA-D; 81.99 for W5-D.

Retrospective BTR.

Replay utility varies across the PSI domain

G · Retain all peptides; WT PSI >5 has no observed positive hits

Sources: [U1] · [D1]

PSI(WT)	Peptides (n)	LoRA-D P@10	W5-D P@10
≤3	91	71.10%	62.64%
(3;4]	85	63.76%	52.59%
(4;5]	36	19.72%	19.44%
>5	23	0.00%	0.00%

Logged computation is not measured wet-lab cost

G · Worker-attempt durations; resumed jobs may omit earlier attempts

Sources: [L3] · [S2]

Local component	Jobs	Recorded hours	Median minutes / job
Probe	9	0.01	0.09
Extract	9	0.96	5.35
Ridge	540	1.42	0.16
Frozen tuning	30	14.07	27.16
LoRA tuning	30	36.01	73.30

The analysis tool has a baseline; extensions remain open

G · Historical software checks do not validate the expanded workflows

Sources: [S3] · [S4] · [S1]

Implemented baseline

FastAPI + React; ETL, API and baseline inference. Serving model: m3-ext4.0-platt-v1.

Current prototype

Workspace UI v2 uses interactive fixtures and tab memory; it is not connected to the backend.

Still required

Projects/history, protein/batch jobs, import/comparison, PostgreSQL acceptance and correct R3 integration.

What do the research questions currently support?

H · Provisional synthesis; favorable and unfavorable results are retained

Sources: [K1] · [K2] · [L1] · [U1]

RQ	Supported evidence	Unresolved questions
1	Broader training helps Ridge, especially at high PSI	Not guaranteed for every head or LoRA arm
2	ESM CNN improves its matched pointwise control	No universal architecture winner established
3	Scanning pretraining helps one-hot CNN on W	Not external validation or ESM+scan transfer
4	LoRA improves matched frozen B/D models	Full tuning, independent validation and cost benefit untested
5	Positive k10 replay; calibration quantified	No LoRA-T, external-source or prospective wet-lab result

Scientific limits accompany every result

H · External / structure inputs remain pending; published results cannot fill them

Sources: [M1] · [L1] · [L2] · [U1] · [S2]

Generalization

Grouping is heuristic. Assays/labs are related; peptide context differs from native proteins.

Model selection

Many recipes have been inspected on development data. Post-hoc CIs do not include new splits or assays.

Practical utility

Replay utility is retrospective. CT553H has no prospective wet-lab, half-life, E3 or measured monetary/time benefit.

Progress conclusions remain open to revision

H · Provisional conclusions; not an acceptance record

Sources: [M1] · [L1] · [U1] · [S1]

Completed work

Traceable labels, QC and splits; Ridge–pLM–CNN–LoRA controls; 88 v3 recipes and an equal-budget replay.

Current support

LoRA improves matched frozen models on W. LoRA-D recovers more positive hits than W5/random in development replay.

Not yet decided

No final production model or overall conclusion. The G2 dossier is ready for review; no closure decision is recorded.

Next steps should answer explicit questions

H · Proposed work; this presentation update runs no training or wet-lab experiment

Sources: [M1] · [L1] · [U1] · [S1]

Review scope

Read evidence + limitations

Record G2 decision

Further evaluation

Lock recipe / RQ

Compatible source

Audit overlap

Serving

Refit / calibrator export + parity acceptance G3

Prospective utility

For utility claims:

Prospective selection

Measure hits/resources

Appendix: retain evidence for review and updates

Reference material; not every appendix slide needs an oral presentation

Sources: [L3]

Complete v3 results

47 Kaggle + 41 local recipes; seed/runtime retained. W MAE, RMSE, AUROC+ and AP+.

Historical controls

R6, R2, P3, merge, binary calibration and domain gates retain their own protocols.

Source lookup

[P]: primary papers; [D/M]: data/code; [K/L/U/H/S]: internal artifacts. Notes and the evidence map provide locators and hashes.

V3 Kaggle: recipe 1–7/47

W: 235 peptides / 221 groups; AUC+/AP+: 184 eligible peptides; runtimes remain separate

Sources: [L3] · [K1]

Recipe	Seed	MAE ↓	RMSE ↓	AUC+ ↑	AP+ ↑
cnn-transfer-dpsi_rep1-D	17,29,43	0.3872	0.5139	0.8425	0.5024
cnn-transfer-dpsi_rep12_mean-D	17,29,43	0.3873	0.5151	0.8391	0.4985
cnn-transfer-scratch-D	17,29,43	0.3936	0.5240	0.8291	0.4597
composition-w5-B	17,29,43	0.4342	0.5488	0.8111	0.4040
composition-w5-D	17,29,43	0.4075	0.5317	0.8109	0.4047
esm2_35m-cnn-B	17,29,43	0.3880	0.5120	0.8506	0.5097
esm2_35m-cnn-D	17,29,43	0.3844	0.5101	0.8488	0.5095

V3 Kaggle: recipe 8–14/47

W: 235 peptides / 221 groups; AUC+/AP+: 184 eligible peptides; runtimes remain separate

Sources: [L3] · [K1]

Recipe	Seed	MAE ↓	RMSE ↓	AUC+ ↑	AP+ ↑
esm2_35m-cnn-E	17,29,43	0.3856	0.5107	0.8474	0.5066
esm2_35m-context-B	17,29,43	0.4281	0.5437	0.8432	0.4924
esm2_35m-context-D	17,29,43	0.4023	0.5229	0.8445	0.4954
esm2_35m-frozen-B	17	0.4071	0.5286	0.8233	0.4484
esm2_35m-frozen-D	17	0.3919	0.5175	0.8221	0.4486
esm2_35m-mcnn-B	17,29,43	0.3855	0.5110	0.8499	0.5036
esm2_35m-mcnn-D	17,29,43	0.3850	0.5109	0.8463	0.5042

V3 Kaggle: recipe 15–21/47

W: 235 peptides / 221 groups; AUC+/AP+: 184 eligible peptides; runtimes remain separate

Sources: [L3] · [K1]

Recipe	Seed	MAE ↓	RMSE ↓	AUC+ ↑	AP+ ↑
esm2_35m-mcnn-E	17,29,43	0.3838	0.5095	0.8470	0.5064
esm2_35m-only-B	17,29,43	0.4158	0.5322	0.8109	0.4291
esm2_35m-only-D	17,29,43	0.4008	0.5217	0.8131	0.4340
esm2_35m-pointwise-B	17,29,43	0.4004	0.5256	0.8375	0.4656
esm2_35m-pointwise-D	17,29,43	0.3897	0.5152	0.8375	0.4739
esm2_35m-pointwise-E	17,29,43	0.3886	0.5150	0.8357	0.4712
nested-family-selection	17,29,43	0.3975	0.5180	0.8367	0.4788

V3 Kaggle: recipe 22–28/47

W: 235 peptides / 221 groups; AUC+/AP+: 184 eligible peptides; runtimes remain separate

Sources: [L3] · [K1]

Recipe	Seed	MAE ↓	RMSE ↓	AUC+ ↑	AP+ ↑
onehot-cnn-B	17,29,43	0.3997	0.5268	0.8304	0.4575
onehot-cnn-D	17,29,43	0.3936	0.5240	0.8291	0.4597
onehot-cnn-E	17,29,43	0.3947	0.5255	0.8267	0.4564
onehot-mcnn-B	17,29,43	0.3985	0.5266	0.8267	0.4516
onehot-mcnn-D	17,29,43	0.3935	0.5242	0.8293	0.4588
onehot-mcnn-E	17,29,43	0.3921	0.5236	0.8289	0.4592
onehot-pointwise-B	17,29,43	0.4059	0.5302	0.8245	0.4299

V3 Kaggle: recipe 29–35/47

W: 235 peptides / 221 groups; AUC+/AP+: 184 eligible peptides; runtimes remain separate

Sources: [L3] · [K1]

Recipe	Seed	MAE ↓	RMSE ↓	AUC+ ↑	AP+ ↑
onehot-pointwise-D	17,29,43	0.3955	0.5238	0.8269	0.4424
onehot-pointwise-E	17,29,43	0.3920	0.5253	0.8235	0.4426
ordered-w3-B	17,29,43	0.4331	0.5492	0.8280	0.4511
ordered-w5-A	17,29,43	0.4711	0.5867	0.8192	0.4436
ordered-w5-B	17,29,43	0.4336	0.5505	0.8312	0.4632
ordered-w5-C	17,29,43	0.4177	0.5372	0.8314	0.4606
ordered-w5-D	17,29,43	0.4078	0.5308	0.8319	0.4614

V3 Kaggle: recipe 36–42/47

W: 235 peptides / 221 groups; AUC+/AP+: 184 eligible peptides; runtimes remain separate

Sources: [L3] · [K1]

Recipe	Seed	MAE ↓	RMSE ↓	AUC+ ↑	AP+ ↑
ordered-w5-D-fraction0.25	17,29,43	0.4154	0.5392	0.8231	0.4468
ordered-w5-D-fraction0.5	17,29,43	0.4113	0.5346	0.8280	0.4525
ordered-w5-D-fraction0.75	17,29,43	0.4083	0.5319	0.8297	0.4555
ordered-w5-E	17,29,43	0.4036	0.5316	0.8275	0.4499
pair-w5-B	17,29,43	0.4149	0.5352	0.8093	0.4024
pair-w5-D	17,29,43	0.4017	0.5263	0.8092	0.4040
sat-only-D	17,29,43	0.4078	0.5308	0.8319	0.4614

V3 Kaggle: recipe 43–47/47

W: 235 peptides / 221 groups; AUC+/AP+: 184 eligible peptides; runtimes remain separate

Sources: [L3] · [K1]

Recipe	Seed	MAE ↓	RMSE ↓	AUC+ ↑	AP+ ↑
sat-quality-D	17,29,43	0.4118	0.5339	0.8307	0.4579
scan-mean12-D	17,29,43	0.4048	0.5276	0.8334	0.4643
scan-rep1-D	17,29,43	0.4047	0.5275	0.8334	0.4643
unordered-w5-B	17,29,43	0.4316	0.5474	0.8260	0.4414
unordered-w5-D	17,29,43	0.4060	0.5298	0.8258	0.4399

V3 Local: recipe 1–7/41

W: 235 peptides / 221 groups; AUC+/AP+: 184 eligible peptides; runtimes remain separate

Sources: [L3] · [L1]

Recipe	Seed	MAE ↓	RMSE ↓	AUC+ ↑	AP+ ↑
esm2_8m-only-B	17,29,43	0.4150	0.5314	0.8107	0.4398
esm2_8m-context-B	17,29,43	0.4288	0.5440	0.8419	0.4943
esm2_8m-only-D	17,29,43	0.3995	0.5206	0.8128	0.4375
esm2_8m-context-D	17,29,43	0.4018	0.5226	0.8433	0.4948
esm2_35m-only-B	17,29,43	0.4157	0.5321	0.8109	0.4290
esm2_35m-context-B	17,29,43	0.4280	0.5436	0.8432	0.4921
esm2_35m-only-D	17,29,43	0.4007	0.5217	0.8131	0.4340

V3 Local: recipe 8–14/41

W: 235 peptides / 221 groups; AUC+/AP+: 184 eligible peptides; runtimes remain separate

Sources: [L3] · [L1]

Recipe	Seed	MAE ↓	RMSE ↓	AUC+ ↑	AP+ ↑
esm2_35m-context-D	17,29,43	0.4022	0.5229	0.8445	0.4954
esm2_150m-only-B	17,29,43	0.4183	0.5346	0.8029	0.4229
esm2_150m-context-B	17,29,43	0.4296	0.5442	0.8381	0.4894
esm2_150m-only-D	17,29,43	0.4031	0.5243	0.8034	0.4258
esm2_150m-context-D	17,29,43	0.4041	0.5241	0.8390	0.4856
esm2_650m-only-B	17,29,43	0.4204	0.5372	0.8033	0.4105
esm2_650m-context-B	17,29,43	0.4319	0.5475	0.8392	0.4832

V3 Local: recipe 15–21/41

W: 235 peptides / 221 groups; AUC+/AP+: 184 eligible peptides; runtimes remain separate

Sources: [L3] · [L1]

Recipe	Seed	MAE ↓	RMSE ↓	AUC+ ↑	AP+ ↑
esm2_650m-only-D	17,29,43	0.4056	0.5281	0.8022	0.4093
esm2_650m-context-D	17,29,43	0.4071	0.5283	0.8394	0.4796
esm1b_650m-only-B	17,29,43	0.4203	0.5380	0.8016	0.4005
esm1b_650m-context-B	17,29,43	0.4295	0.5462	0.8359	0.4773
esm1b_650m-only-D	17,29,43	0.4054	0.5283	0.8024	0.4088
esm1b_650m-context-D	17,29,43	0.4069	0.5289	0.8360	0.4752
esm1v_650m_1-only-B	17,29,43	0.4179	0.5347	0.8022	0.4153

V3 Local: recipe 22–28/41

W: 235 peptides / 221 groups; AUC+/AP+: 184 eligible peptides; runtimes remain separate

Sources: [L3] · [L1]

Recipe	Seed	MAE ↓	RMSE ↓	AUC+ ↑	AP+ ↑
esm1v_650m_1-context-B	17,29,43	0.4289	0.5435	0.8384	0.4838
esm1v_650m_1-only-D	17,29,43	0.4032	0.5247	0.8027	0.4194
esm1v_650m_1-context-D	17,29,43	0.4050	0.5252	0.8378	0.4814
esmc_300m-only-B	17,29,43	0.4203	0.5386	0.8001	0.4038
esmc_300m-context-B	17,29,43	0.4294	0.5455	0.8379	0.4806
esmc_300m-only-D	17,29,43	0.4046	0.5279	0.8011	0.4062
esmc_300m-context-D	17,29,43	0.4038	0.5250	0.8391	0.4820

V3 Local: recipe 29–35/41

W: 235 peptides / 221 groups; AUC+/AP+: 184 eligible peptides; runtimes remain separate

Sources: [L3] · [L1]

Recipe	Seed	MAE ↓	RMSE ↓	AUC+ ↑	AP+ ↑
prott5_encoder-only-B	17,29,43	0.4211	0.5385	0.8036	0.4143
prott5_encoder-context-B	17,29,43	0.4301	0.5450	0.8414	0.4874
prott5_encoder-only-D	17,29,43	0.4056	0.5282	0.8049	0.4223
prott5_encoder-context-D	17,29,43	0.4055	0.5259	0.8425	0.4900
ankh_base-only-B	17,29,43	0.4188	0.5349	0.8075	0.4350
ankh_base-context-B	17,29,43	0.4279	0.5423	0.8426	0.4949
ankh_base-only-D	17,29,43	0.4048	0.5265	0.8088	0.4396

V3 Local: recipe 36–41/41

W: 235 peptides / 221 groups; AUC+/AP+: 184 eligible peptides; runtimes remain separate

Sources: [L3] · [L1]

Recipe	Seed	MAE ↓	RMSE ↓	AUC+ ↑	AP+ ↑
ankh_base-context-D	17,29,43	0.4035	0.5231	0.8440	0.4988
nested-family-selection	17,29,43	0.3945	0.5139	0.8409	0.4877
esm2_35m-frozen-B	17,29,43	0.4034	0.5243	0.8263	0.4554
esm2_35m-frozen-D	17,29,43	0.3904	0.5155	0.8246	0.4544
esm2_35m-lora-B	17,29,43	0.3637	0.4810	0.8550	0.5517
esm2_35m-lora-D	17,29,43	0.3650	0.4842	0.8536	0.5474

H-sat: frozen ESM and tile-score pretraining

Appendix · H-sat test: 91 peptides / 89 groups / 47,847 cells

Sources: [H4] · [H6] · [H9]

Recipe	AUROC+	Limitation
R2 M3 / M3+ESM / ESM	0.8208 / 0.8230 / 0.7853	ESM2-650M + PCA64/Ridge; different from v3 CNN heads
P3 ext ridge / random / pretrained	0.8208 / 0.8175 / 0.8098	Tested Voutsinos pretraining did not establish a benefit
Merge sat / scan-only	0.8208 / 0.7277	Different populations do not isolate a source effect
Merge naive / source / prior	0.8189 / 0.8203 / 0.8197	Pooling did not outperform the saturation Ridge control
Merge neural random / pretrain	0.8154 / 0.8168	Separate from the newer stage06 one-hot CNN

Scanning M0–M4: context and population transfer

Appendix · H-scan: mean metric across seeds; H-transfer: saved score on 91 peptides

Sources: [H10]

Representation	H-scan AUROC	H-transfer AUROC
M0_composition_only	0.8393	0.7071
M1_plus_position_index	0.8445	0.7194
M2_plus_window_composition	0.8625	0.7241
M3_plus_ordered_context	0.8682	0.7273
M4_context_and_position	0.8684	0.7253

G2 saturation retains every gate / PAP control

Appendix · 91 peptides / 89 groups; public Degformer has training exposure

Sources: [H1]

Recipe	AUROC+	Scope
DeltaPAP	0.769284	H-sat; development
ridge	0.808186	H-sat; development
gate_pap	0.800225	H-sat; development
no_gate_pap	0.793727	H-sat; development
gate_no_pap	0.801510	H-sat; development
Degformer	0.924814	Checkpoint overlap

R6: recipe 1–7/23

Appendix · Fixed legacy folds; not directly comparable with full-PSI W

Sources: [H5]

Recipe	AUROC+ ensemble	Scope
esm150_mcnn	0.821244	H-sat; 3 seeds × 5 folds
esm35_cnn1	0.826394	H-sat; 3 seeds × 5 folds
esm35_mcnn	0.830490	H-sat; 3 seeds × 5 folds
esm35_mcnn+scan_ft	0.829874	H-sat; 3 seeds × 5 folds
esm35_mcnn+scan_pool	0.836670	H-sat; 3 seeds × 5 folds
esm650_mcnn	0.824286	H-sat; 3 seeds × 5 folds
esm8_mcnn	0.826500	H-sat; 3 seeds × 5 folds

R6: recipe 8–14/23

Appendix · Fixed legacy folds; not directly comparable with full-PSI W

Sources: [H5]

Recipe	AUROC+ ensemble	Scope
oh_cnn1	0.812747	H-sat; 3 seeds × 5 folds
oh_mcnn	0.816805	H-sat; 3 seeds × 5 folds
oh_mcnn+scan_ft	0.831466	H-sat; 3 seeds × 5 folds
oh_mcnn+scan_pool	0.824152	H-sat; 3 seeds × 5 folds
ridge_M3	0.820822	H-sat; 3 seeds × 5 folds
ridge_W10	0.818231	H-sat; 3 seeds × 5 folds
ridge_W5	0.828637	H-sat; 3 seeds × 5 folds

R6: recipe 15–21/23

Appendix · Fixed legacy folds; not directly comparable with full-PSI W

Sources: [H5]

Recipe	AUROC+ ensemble	Scope
ridge_W7	0.825471	H-sat; 3 seeds × 5 folds
ridge_esm150	0.820682	H-sat; 3 seeds × 5 folds
ridge_esm35	0.823387	H-sat; 3 seeds × 5 folds
ridge_esm650	0.823038	H-sat; 3 seeds × 5 folds
ridge_esm8	0.821393	H-sat; 3 seeds × 5 folds
ridge_scan_pool	0.813758	H-sat; 3 seeds × 5 folds
ridge_scan_pool_t3	0.817291	H-sat; 3 seeds × 5 folds

R6: recipe 22–23/23

Appendix · Fixed legacy folds; not directly comparable with full-PSI W

Sources: [H5]

Recipe	AUROC+ ensemble	Scope
ridge_scan_prior	0.819041	H-sat; 3 seeds × 5 folds
ridge_scan_prior_t3	0.819173	H-sat; 3 seeds × 5 folds

R6 primary and contrast families retain separate verdicts

Appendix · A secondary / G-family gain does not replace the primary verdict

Sources: [H5]

Contrast	ΔAUROC	Interval used for inference	Status
P: ESM35 mCNN–M3	+0.00967	[−0.00252; +0.02326]	Primary unresolved
A: W5–W3	+0.00781	[+0.00212; +0.01392]	Supported; family A, m=3
S1: ESM–one-hot mCNN	+0.01369	[+0.00253; +0.02617]	Positive secondary contrast
S6: mCNN–CNN	+0.00410	[−0.00050; +0.00868]	Unresolved
G: one-hot scan FT–scratch	+0.01466	[+0.00272; +0.02759]	Supported; family G, m=4

P6 and R3 evaluate different endpoints

Appendix · Binary Brier values are not directly comparable with three-class v3 Brier

Sources: [H7] · [H8]

Experiment / recipe	Binary Brier ↓	AUROC or ECE	Endpoint
P6 raw	0.1272	ECE 0.1437	P(ΔPSI≥1), H-sat
P6 constant	0.1139	ECE 0.0154	P(ΔPSI≥1), H-sat
P6 Platt	0.0925	ECE 0.0163	P(ΔPSI≥1), H-sat
P6 isotonic	0.0925	ECE 0.0164	P(ΔPSI≥1), H-sat
R3 tier1 domain gate	0.0915	AUC 0.9111	WT PSI≤4; 235 peptides
R3 tier2 stageB	0.1434	AUC 0.8154	WT PSI≤4; 222 groups

Additional transfer and training diagnostics

Appendix · Replacement-pair cohorts and selected epochs remain development analyses

Sources: [K2]

Cohort substitution	Cell count	Scratch MAE	Pretrain MAE
20 scanning-type pairs	6,433	0.3991	0.3913
360 other pairs	115,979	0.3933	0.3870

CNN / pointwise-D selects epoch 1 in 9/15 fits; mCNN-D median = 3.

Raising the epoch cap has no demonstrated benefit; k1 / k5 heads have 36,833 / 57,185 parameters.

Each LoRA seed improves its matched frozen control

Appendix · W peptide-macro MAE; seeds share the outer partition

Sources: [L1]

Recipe	Seed17	Seed29	Seed43	Ensemble
frozen-B	0.4073	0.4031	0.4059	0.4034
frozen-D	0.3920	0.3918	0.3938	0.3904
lora-B	0.3794	0.3786	0.3779	0.3637
lora-D	0.3794	0.3778	0.3788	0.3650

Replay: recall, negative events and tie sensitivity

Appendix · Secondary analyses do not reselect the primary recipe

Sources: [U1]

Policy	Recall@10 +	P@10 –	BTR50 +	BTR80 +
LoRA-D	16.62%	14.00%	71.32	154.36
W5-D	13.41%	4.34%	81.99	170.53
Random	1.93%	1.33%	255.51	404.48

Positive recall / BTR: 198 hit-containing peptides. Negative precision: all 235.

No k10 cutoff ties; 100 hash seeds preserve Precision@10.

References / verifiable sources 1

Full DOI / URL / locators; internal evidence includes repository paths

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data/processed/v3-fullpsi/tables/saturation-cells.parquet

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[D2] CT553H. Dataset build v3. counts, fold strata, hit rules; historical specifications remain in DATASET.md.

research/kaggle-roadmap-2026-10-31/dataset-build.json

[D3] CT553H. QC / WT references. load_psi, build_saturation, build_scanning.

research/residue-task-derisk-2026-10-15/residue_data.py

[D4] CT553H. Six-bin PSI. bin normalization and weighted index.

research/full-data-audit/build_psi.py

[M1] CT553H. Execution protocol v3.1. arms, fits, hyperparameters, metrics, stages.

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[M2] CT553H. Sequence features. features, crop_inputs; W5 has 260 dimensions.

research/kaggle-roadmap-2026-10-31/degron_v3/features.py

[M3] CT553H. Convolutional heads. ConvHead, fit_head, transfer_stage.

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[M4] CT553H. Frozen / LoRA. LoRALinear, MutationRegressor; rank8 alpha16.

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[K1] CT553H. Kaggle v3 (02/11). recipes, contrasts; run 369cb2fe6a1e31e0.

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[K2] CT553H. Kaggle diagnostics. learning_curve, pair_diagnostics, fold_mae, selected_epochs.

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research/kaggle-roadmap-2026-10-31/local-output-analysis.json

[L2] CT553H. OOF audit / residuals. core_diagnostics, matched_lora_minus_frozen; saved audit.

research/progress-review-2026-11-06/local-audit.json

[L3] CT553H. Output review / recorded computation. 88 recipes, timing and inventory; durations are not GPU-active time.

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[U2] CT553H. Protocol replay. equal budget k10, analytic random, grouped bootstrap.

research/oof-prioritization-2026-11-07/PROTOCOL.md

[H1] CT553H. G2 saturation (18/10). H-sat; DeltaPAP, ridge, gate and checkpoint overlap.

research/g2-saturation-2026-10-18/results.json

[H2] CT553H. P0/P1 learning curve. H-sat train PSI threshold, same test.

research/learning-curve-2026-10-22/results.json

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[H3] CT553H. R1 cleaned Degformer. 3 seeds × 5 folds; paired comparison intervals.

research/r1-degformer-clean-2026-10-22/run-kaggle-20261022/results.json

[H4] CT553H. R2 frozen ESM. PCA64/Ridge ESM2-650M, restricted H-sat.

research/r2-esm-features-2026-10-22/run-kaggle-20261022/results.json

[H5] CT553H. R6 ladder. 23 recipes; primary vs secondary vs G families.

research/r6-ladder-scan-2026-10-23/run-kaggle-20261023/results.json

[H6] CT553H. P3 pretraining. tile-score pretraining; ext/ridge/rand/pre.

research/pretrain-p3-2026-10-22/results.json

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[H7] CT553H. P6 calibration. binary event ΔPSI≥1; different from multiclass v3 Brier.

research/calibration-p6-2026-10-22/results.json

[H8] CT553H. R3 domain gate. domain WT PSI≤4; different endpoint.

research/r3-domain-gate-2026-10-22/run-kaggle-20261022/results.json

[H9] CT553H. Scan–sat merge. pooling/source/prior and neural transfer controls.

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[H10] CT553H. Scanning Ridge M0–M4. scanning seed-mean vs held saturation value; populations differ.

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[S1] CT553H. G2 acceptance dossier. review-ready development; no recorded closure decision.

research/g2-acceptance-2026-11-07/evidence.json

[S2] CT553H. Wet-lab source review. external biological precedents; no direct CT553H wet-lab cost evidence.

research/wet-lab-evidence-plan-2026-11-07/results.json

[S3] CT553H. Serving baseline. baseline scope and historical software checks; serving m3-ext4.0-platt-v1.

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[S4] CT553H. Workspace UI prototype. synthetic fixtures; not connected to backend.

system/design/workspace-ui-v2/README.md